

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married	
	Yes	No
Bachelors or higher	187 (166)	113 (194)
Some college	162 (166)	138 (194)
High school or less	149 (166)	151 (194)

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H_0 : 兩者無關

H_a : 兩者有關

$$\frac{300 \cdot 498}{900} = 166$$

$$\frac{300 \cdot 402}{900} = 134$$

$$\chi^2 = \frac{(187-166)^2}{166} + \frac{(113-166)^2}{166} + \frac{(162-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(138-134)^2}{134} + \frac{(151-134)^2}{134}$$

$$= 10.061$$

$$df = 2, \alpha = 0.01 \quad \chi^2_{2,0.01} = 9.21$$

$$\because 10.061 > 9.21$$

$\therefore$ reject H_0 , 有顯著差異

12, H_0 : 三種住宅的分佈相同

H_a : 三種住宅的分佈不同

$$\chi^2 = 36.499$$

$$\chi^2_{0.01,6} = 16.812$$

$\therefore$ reject H_0 , 有顯著差異

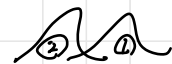
12. **Dissolved Oxygen Content** Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	4.8	5.2	5.0	4.9	5.1
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a. Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.

b. Use a Student's t -test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

H_0 : 兩組且分佈相同

H_a : $P_1 > P_2$ 

$$T_1 = 34 \quad T_1^* = 1 \cdot (5+5+1) - 34 = 21$$

$$n_1 = 5, n_2 = 5, \alpha = 0.05 = 19$$

$$21 > 19$$

$\rightarrow$ non reject

13. $H_0: H = D$

$$T_1 = 45$$

$H_a: H \neq D$

$$T_1^* = 9(9+9+1) - 45 = 126$$

$$\min(T_1, T_1^*) = 45$$

查 $\alpha = 0.05, n_1 = 9, n_2 = 9$, 臨界值 $= 62$

$45 < 62, \rightarrow$ reject

15.2

$H_0: P(A > B) = 0.5$

$H_a: P(A > B) \neq 0.5$

$$2 \cdot [C_0^7 \cdot 0.5^7 + C_1^7 \cdot 0.5^7]$$

$$= 0.125 > 0.1$$

$\rightarrow$ non reject

11. Gourmet Cooking

$\oplus: 11 \quad \ominus: 9$

$$a: X \sim \text{Bin}(20, 0.5)$$

$$P(X \leq 4) = 0.0091$$

$$P(X \leq 5) = 0.0207 \rightarrow 5 < 11 < 15 \rightarrow \text{Non reject}$$

$$b. \quad Z = \frac{X - \mu}{\sigma} = \frac{11 - 20 \cdot 0.5}{\sqrt{20 \cdot 0.5 \cdot 0.5}} = 0.0441 < Z_{0.05} \rightarrow \text{non reject}$$

13.

$$a. \quad +: 2 \quad -: 8 \quad n=10$$

$$H_0: p = 0.5$$

$$H_1: p \neq 0.5$$

$$2(P(X=0) + P(X=1) + P(X=2))$$

$$= 0.11 > 0.05 \quad \text{reject } H_0$$